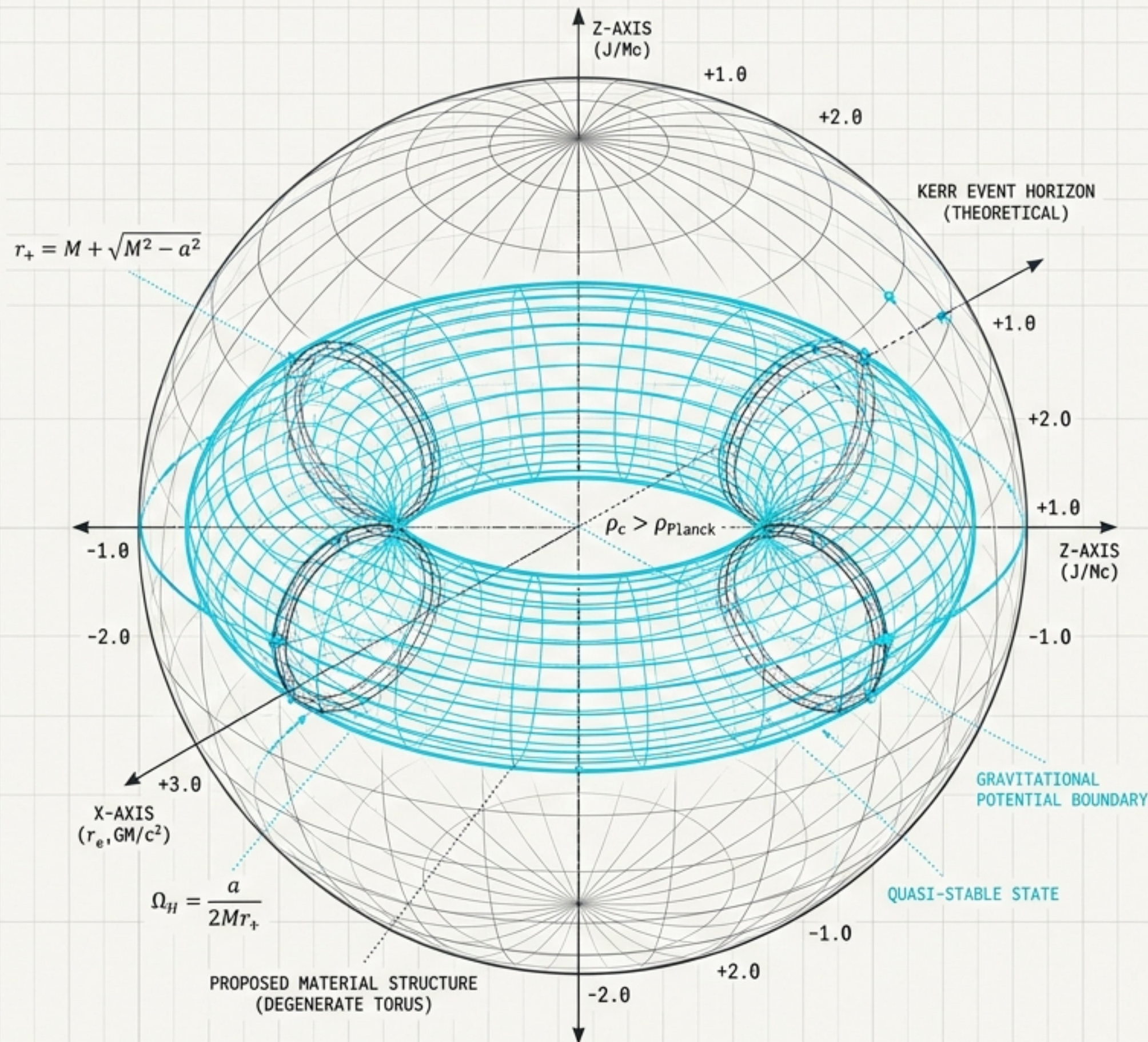


DEGENERACY BEFORE COLLAPSE

A Falsifiable Horizonless Alternative to Black Hole Formation

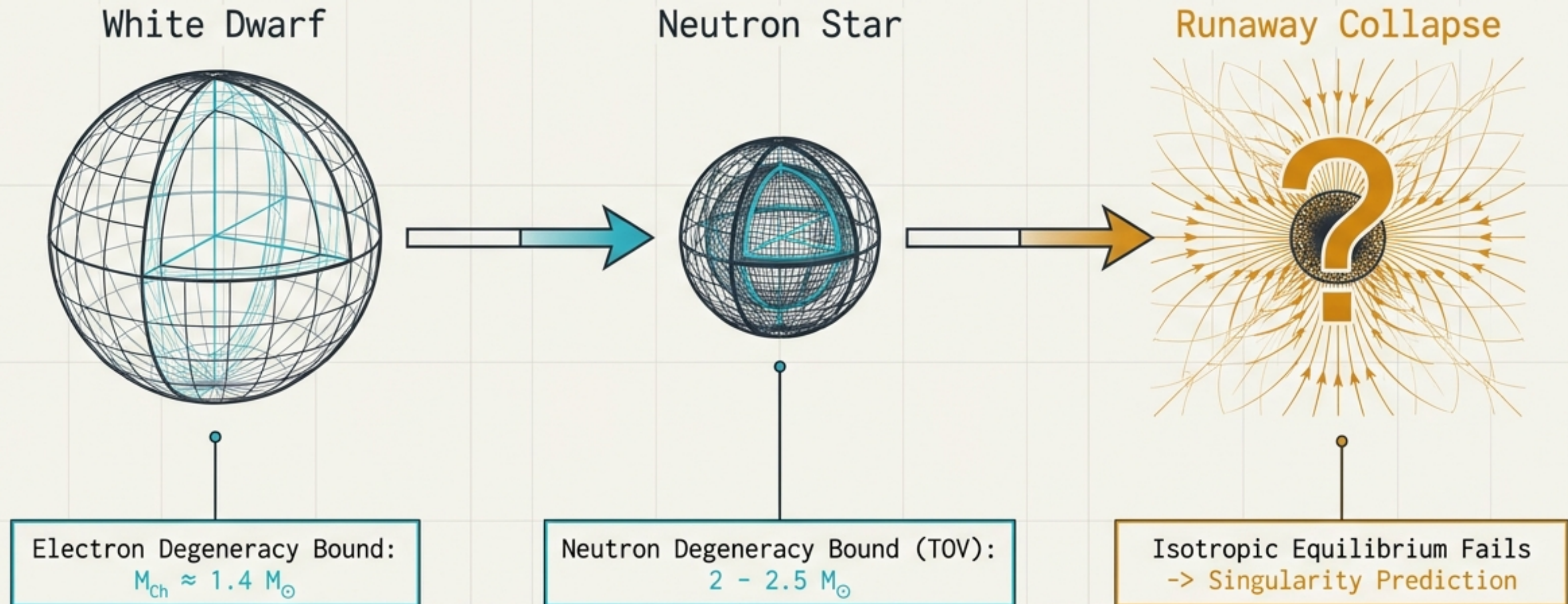
Flyxion, Independent Researcher



The Limits of Known Degeneracy

The Standard Collapse Sequence

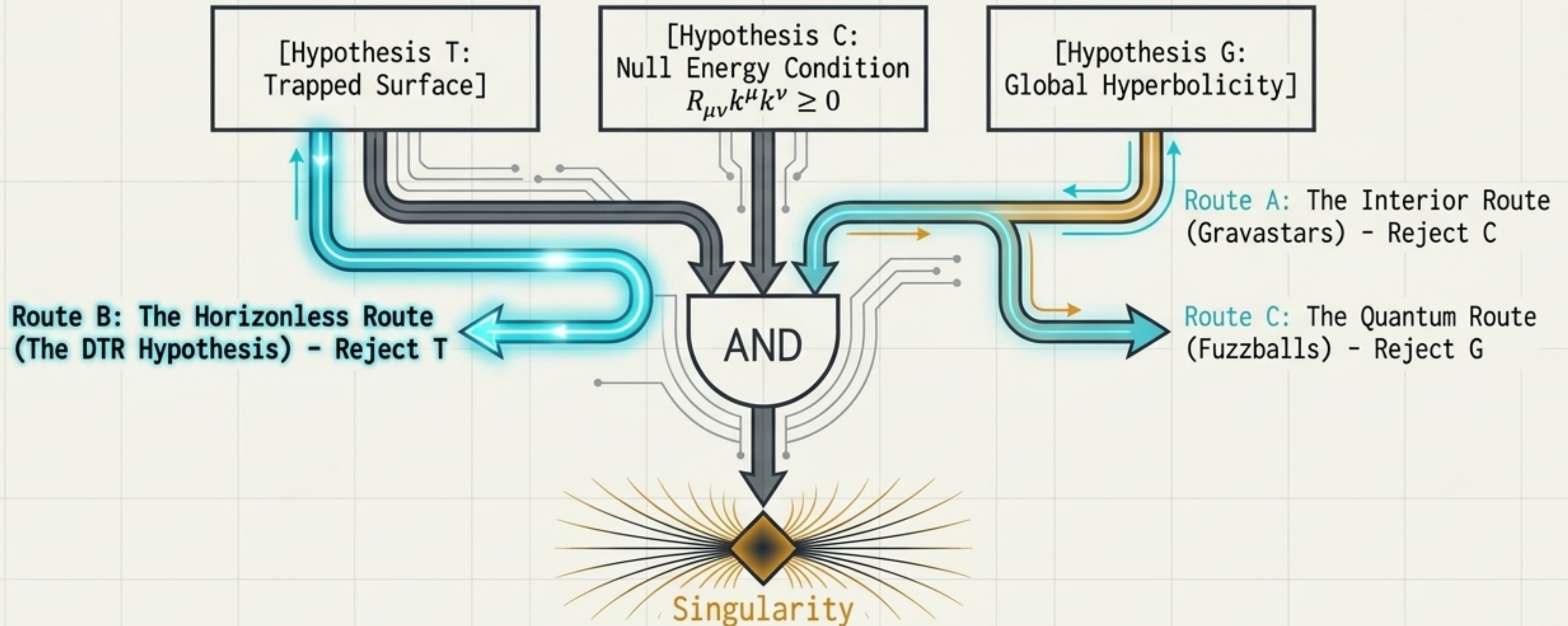
The limits of known degeneracy pressure force massive remnants past the TOV bound. Without a new mechanism, isotropic fluid models invariably predict runaway collapse.



The Penrose Theorem & The Escape Map

A theorem is only as absolute as its hypotheses. Escaping the singularity requires breaking a specific logical gate. We commit to Route B: preventing the trapped surface from forming at all.

Raychaudhuri Focusing: $d\theta/d\lambda \leq -1/2\theta^2$



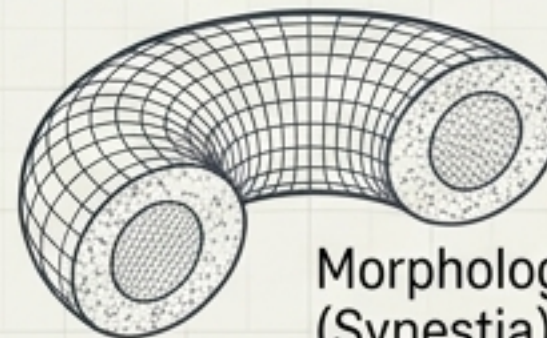
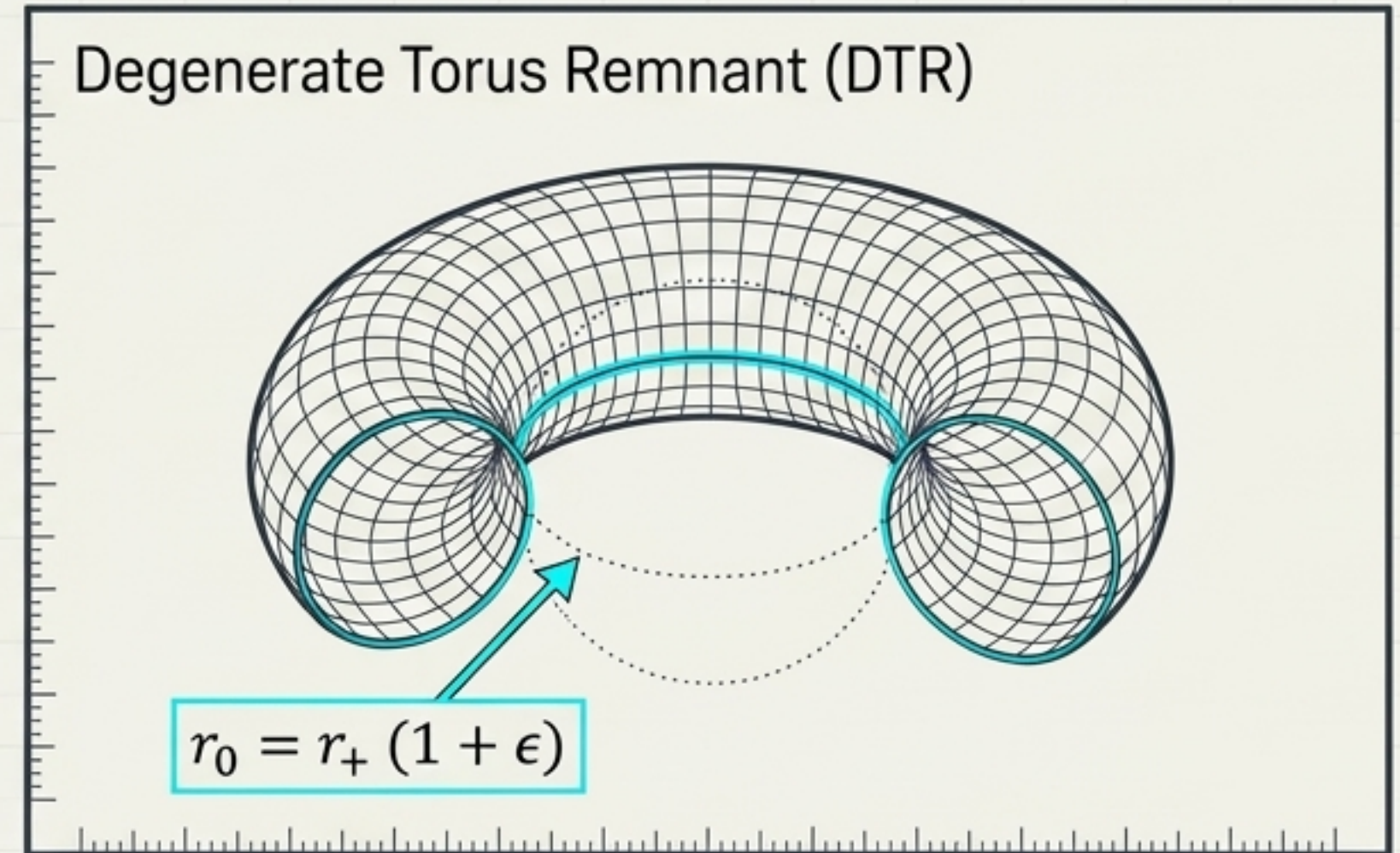
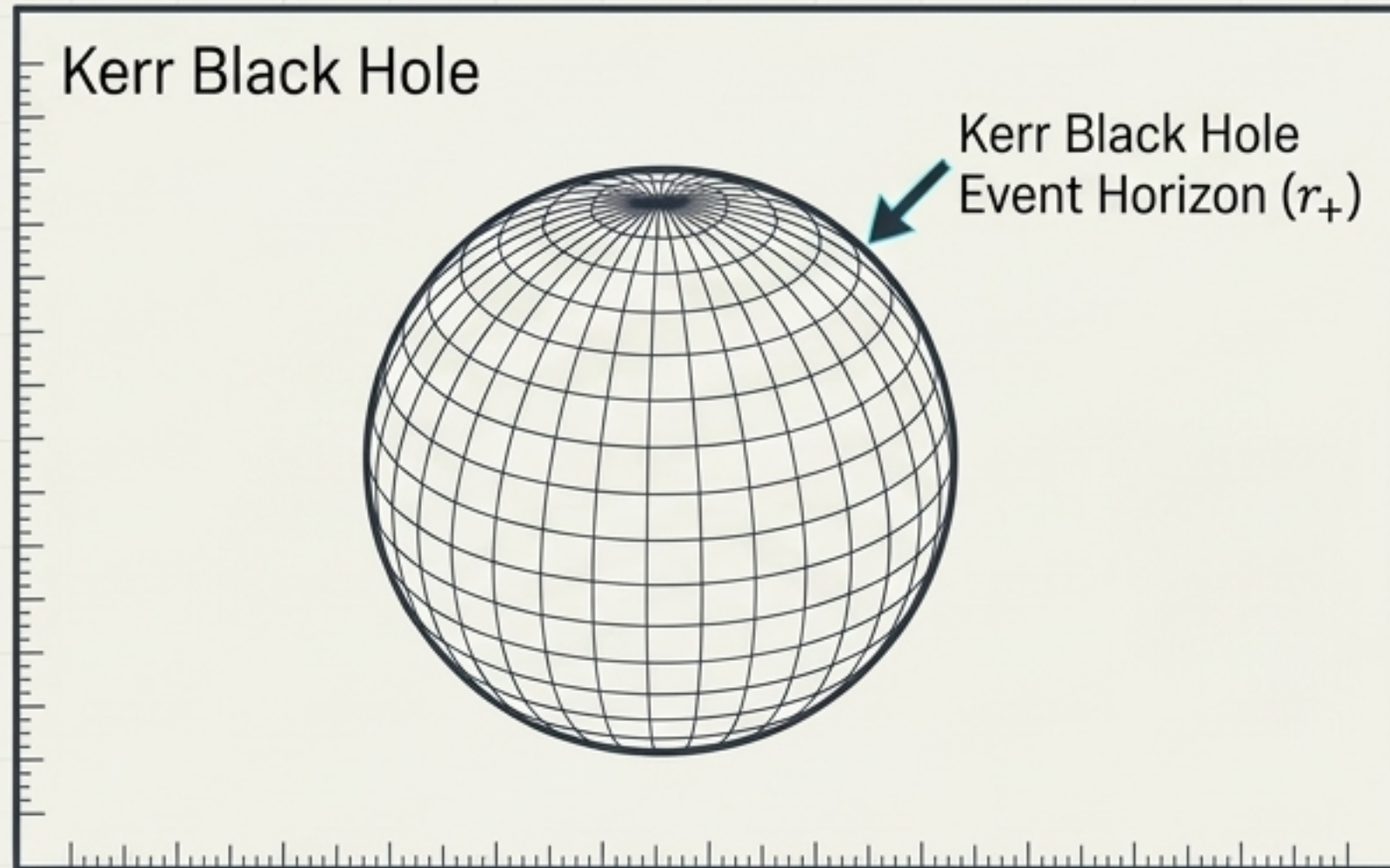
Exotic Compact Object (ECO) Diagnostics

Unlike **gravastars**, a DTR maintains ordinary energy conditions in the bulk.
Unlike black holes, it maintains causal connection with the exterior universe.

	Standard Black Hole	Gravastar	Degenerate Torus Remnant (DTR)
Trapped Surface / Event Horizon Formed?	✓	✓	✗
Null Energy Condition (NEC) Violated?	✗	✓	✗
Interior is Causally Sealed?	✓	✓	✗
Causally Probed via GW Echoes?	✗	✗	✓

Defining the Degenerate Torus Remnant

A DTR is a rapidly rotating configuration of degenerate matter. It halts collapse at a material boundary r_0 , a microscopic proper distance outside the would-be Kerr horizon, $0 < \epsilon \ll 1$.

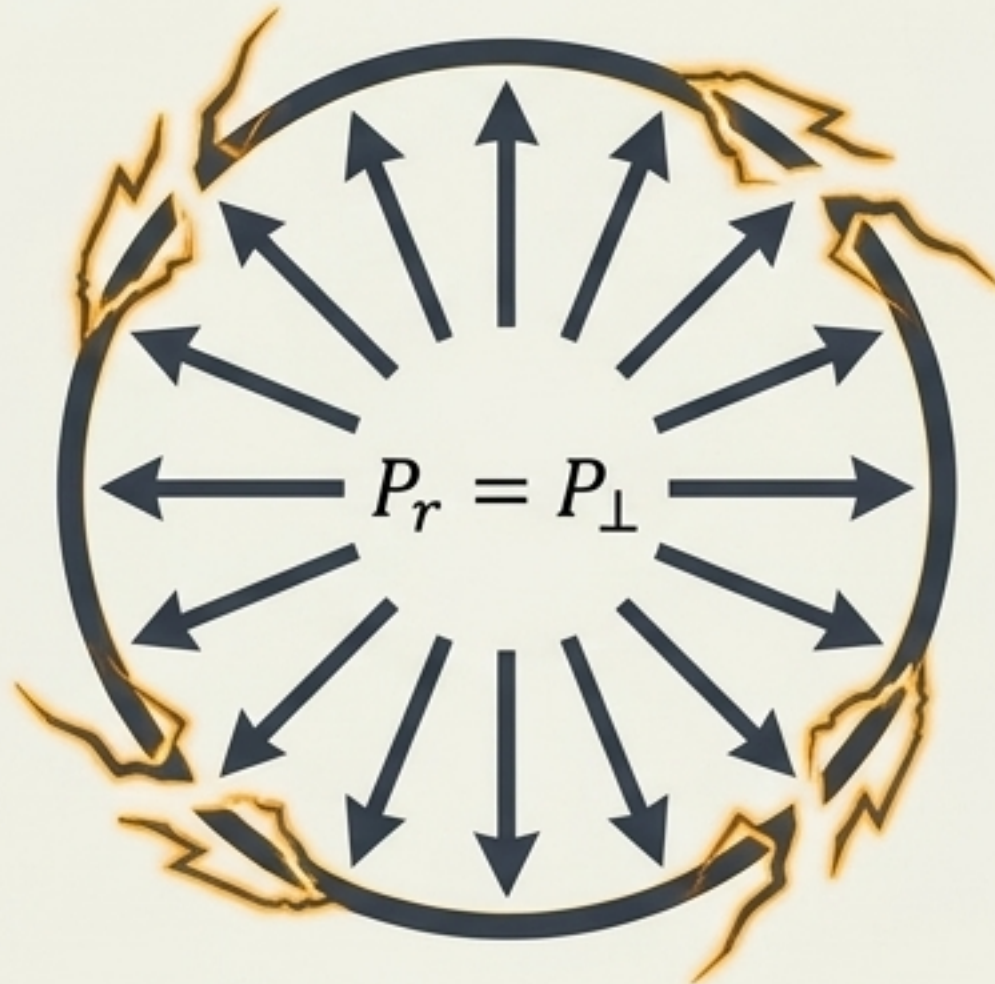


Morphological Analogy: Centrifugal reorganization (Synestia), not a shared equation-of-state.

The Physics of Extreme Compactness

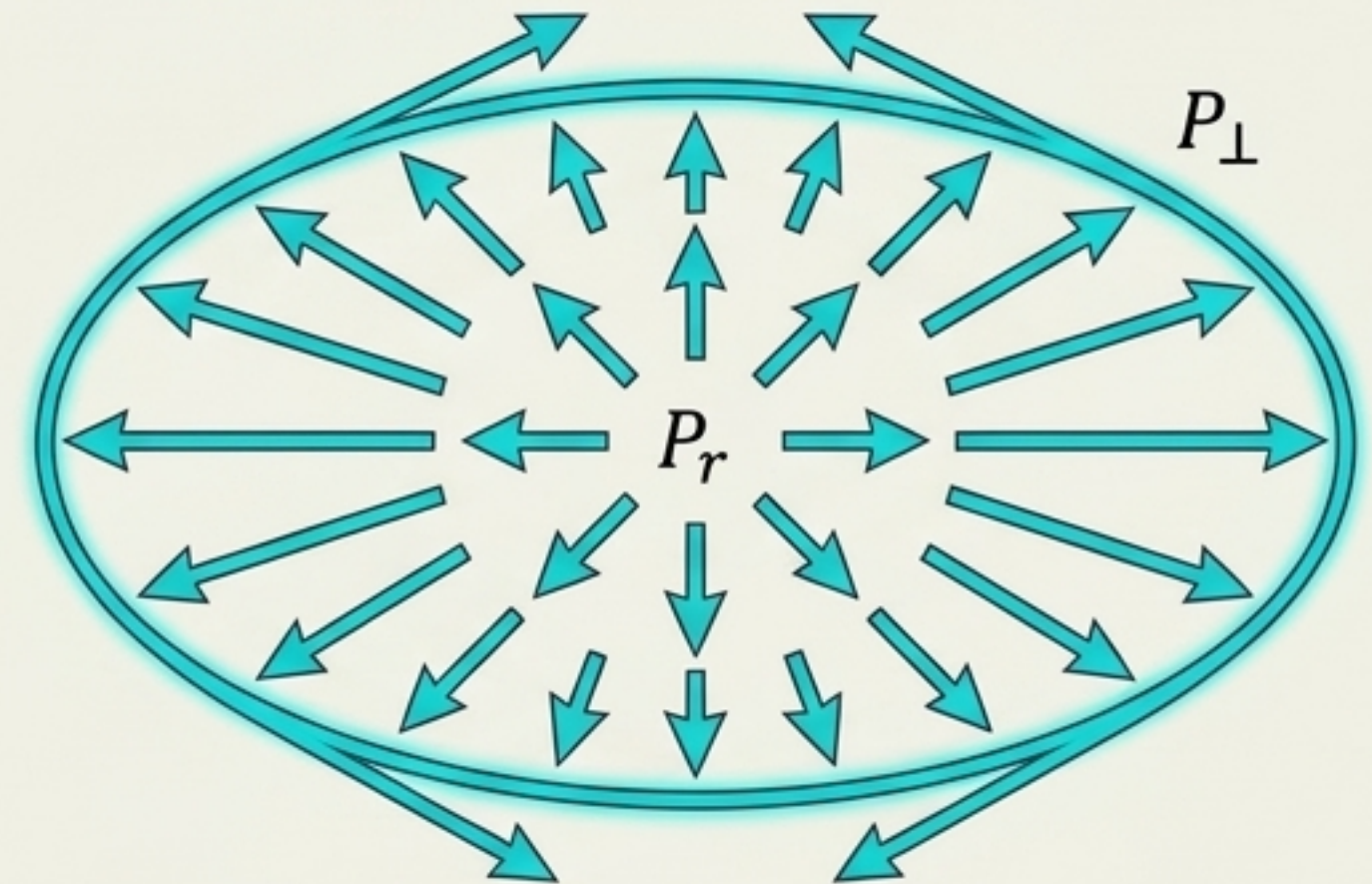
How do we exceed the isotropic $C \leq 8/9$ Buchdahl bound without exotic negative energy? Pressure anisotropy. Centrifugal reorganization into a torus naturally supplies the $P_{\perp} > P_r$ gradient required to support ultra-compact matter.

Isotropic - The Buchdahl Bound



Fails at Compactness $C = 8/9$

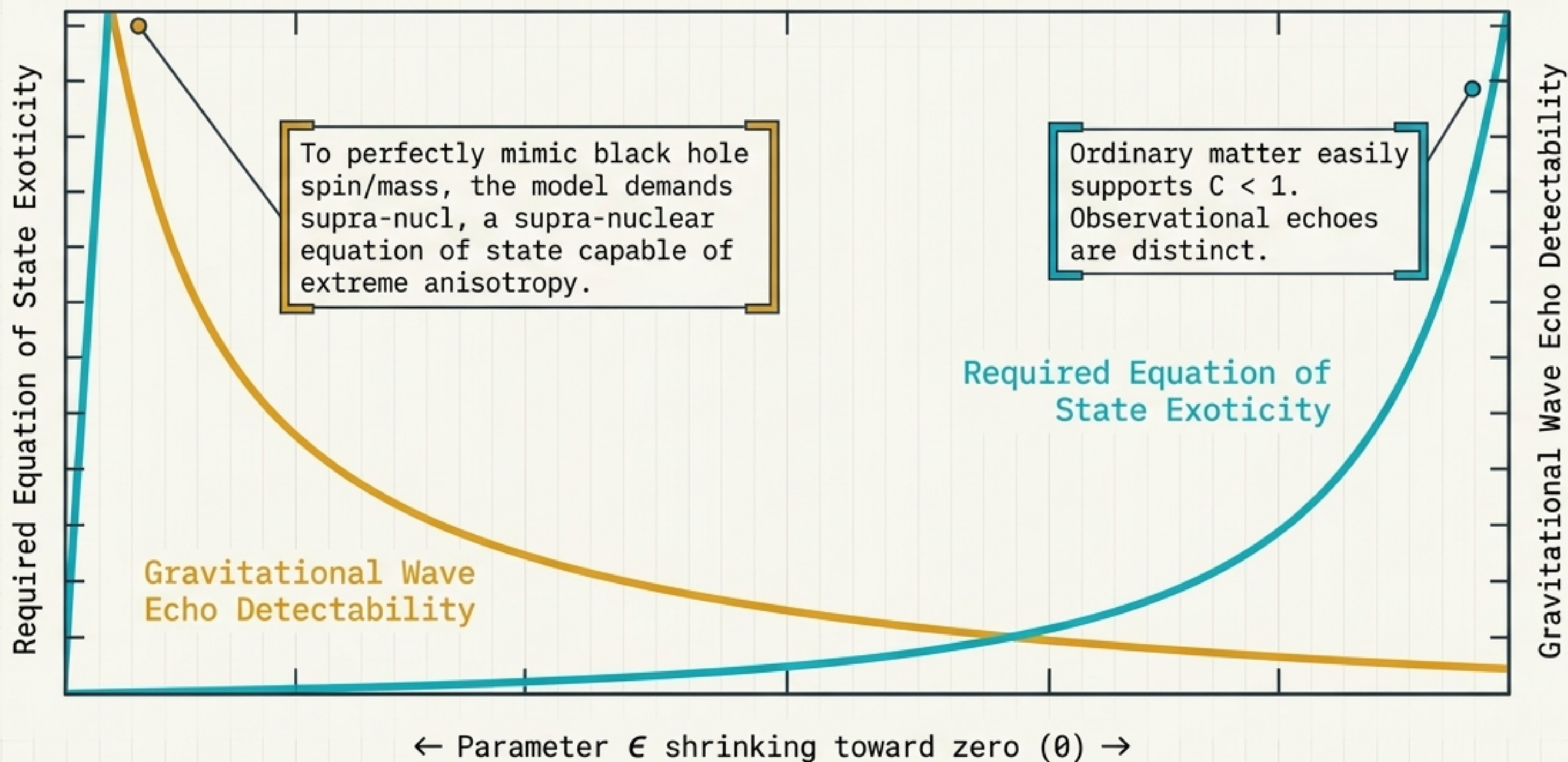
Anisotropic - Bowers-Liang 1974



$P_{\perp} > P_r$: Supports Compactness $C \rightarrow 1$ without violating known energy conditions.

The “Exotic” Boundary: A Clarity Check

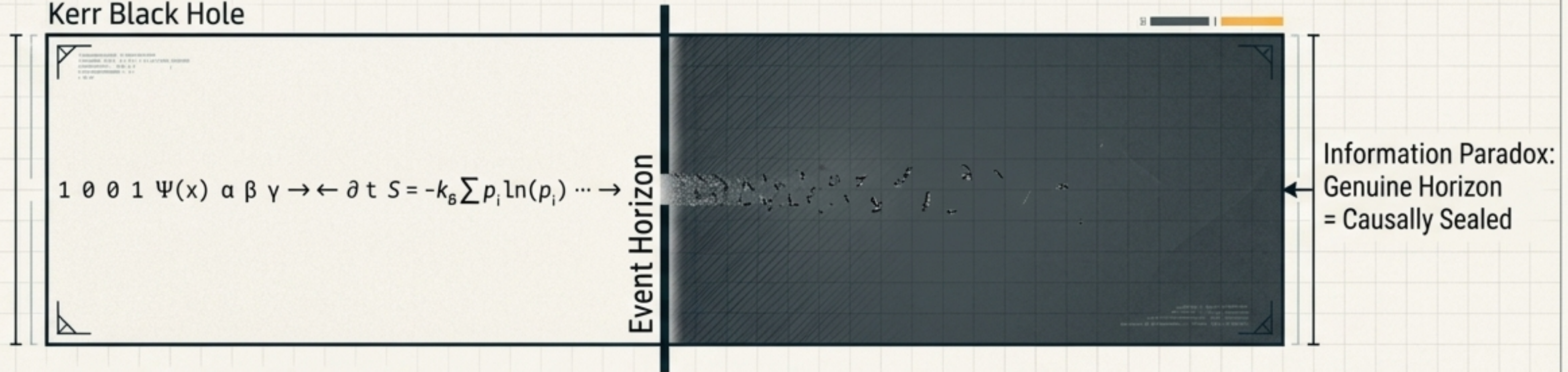
The Trade-off: The regime where echoes are observationally distinct is the exact regime where the required pressure anisotropy is most physically strained.



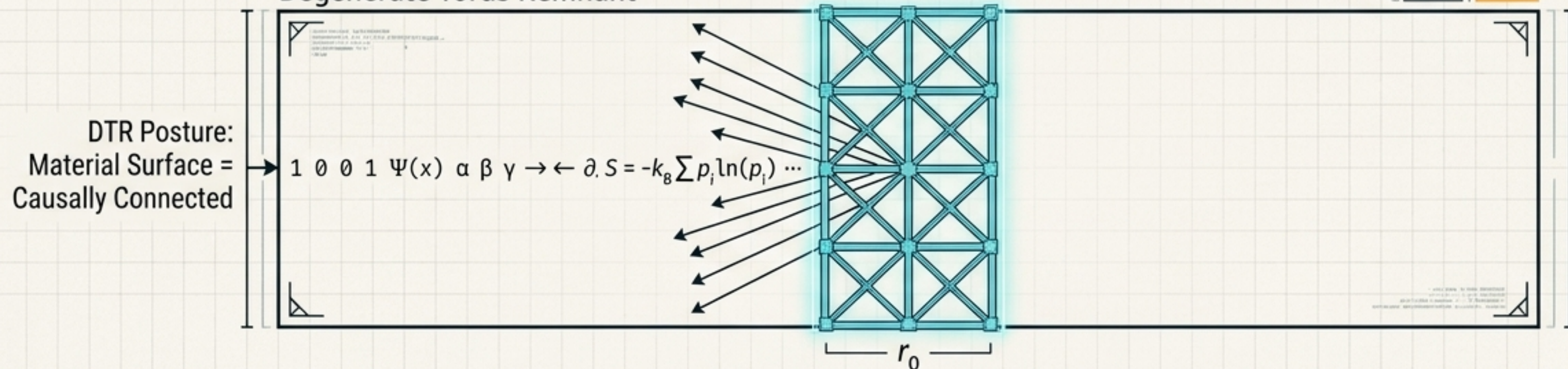
The Recoverability Posture

The DTR does not solve the Information Paradox; it prevents it from arising. Because no horizon ever forms, the remnant's material content remains causally connected to the exterior universe throughout its history.

Kerr Black Hole



Degenerate Torus Remnant



The Open Program: A Blueprint, Not a Solution

The metric and stress-energy are an explicit, well-posed specification for future numerical relativity—not a solved ansatz.

- Formulate stationary metric:

$$ds^2 = -N^2 dt^2 + A^2(d\varpi^2 + dz^2) + B^2\varpi^2(d\phi - \omega dt)^2$$

- Prove horizonlessness:

$$N_{\min} \equiv \inf N(\varpi, z) > 0$$

- Solve field equations as a residual target:

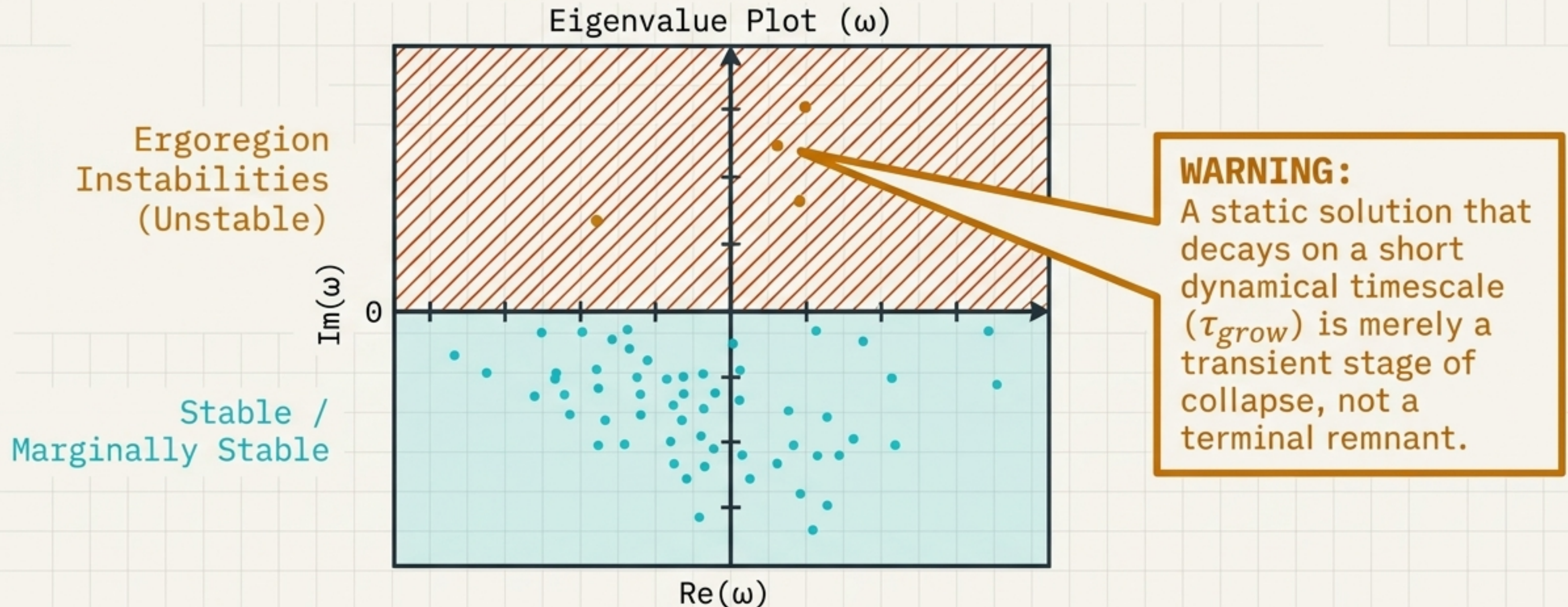
$$\|E[g, T]\|_L^2(\Sigma) < \epsilon_{\text{num}}$$

- Ensure timelike boundary matching via Israel junction conditions: $[K_{ab}]_{\Sigma_0}$

The Final Boss: Dynamical Stability

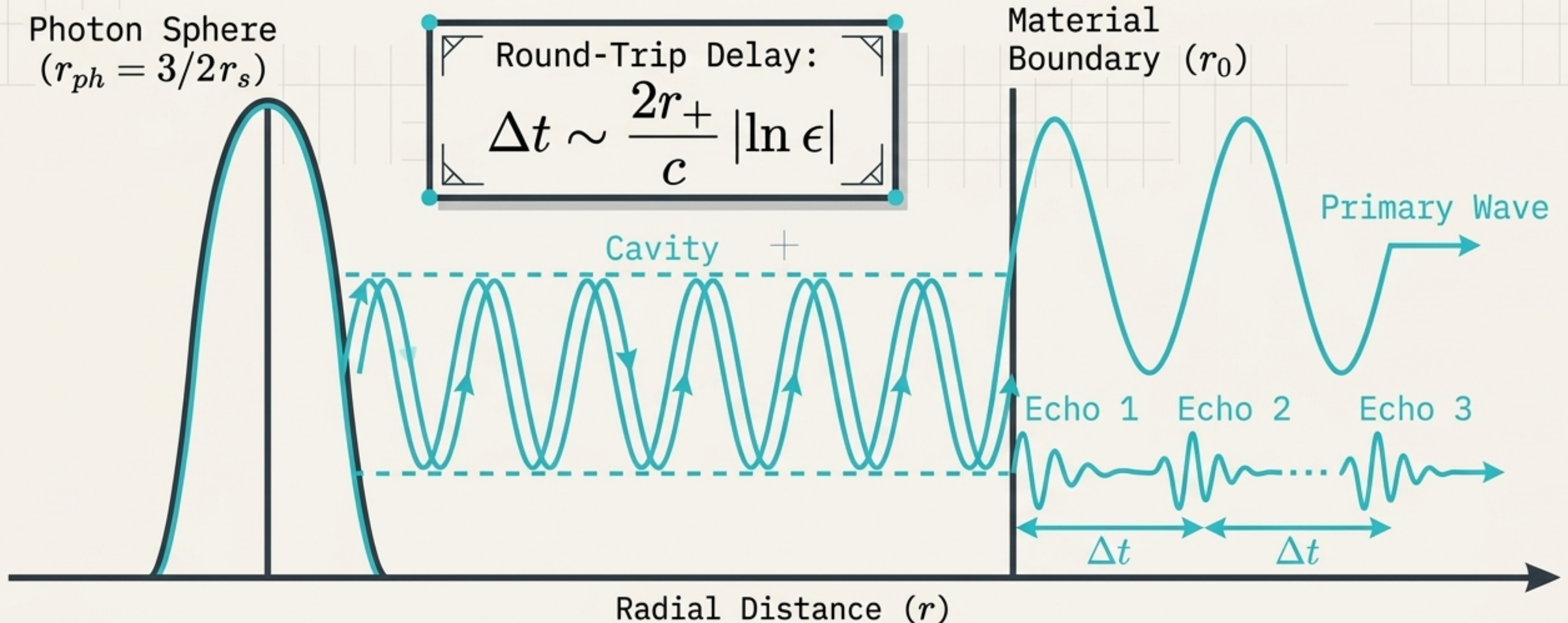
Rotating horizonless objects are uniquely susceptible to ergoregion instabilities. Proving linear stability is the ultimate test for the DTR as a physical endpoint.

Linear perturbation analysis: $L[\delta g, \delta T] = \omega^2 M[\delta g, \delta T]$



The Echo Test: Observational Signatures

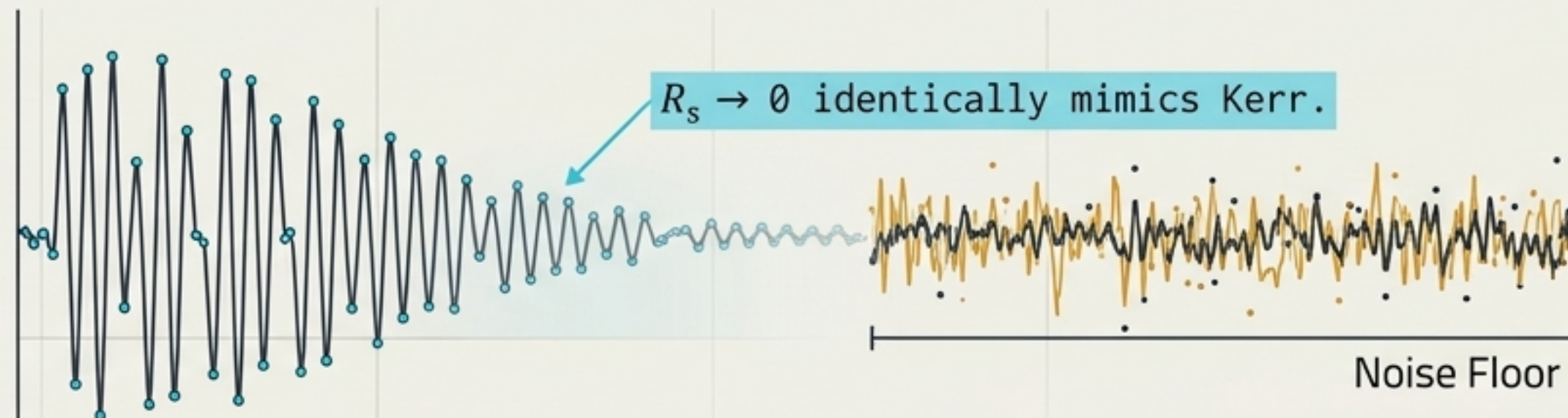
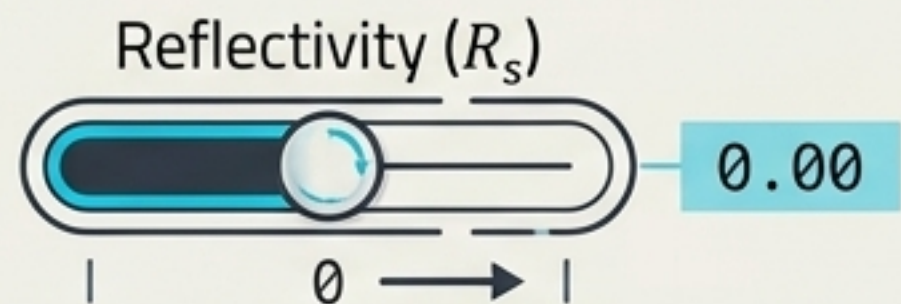
Incident gravitational waves reflect off the DTR's material surface. This produces a train of late-time, repeating echoes following a post-merger ringdown, delayed by the cavity light-crossing time.



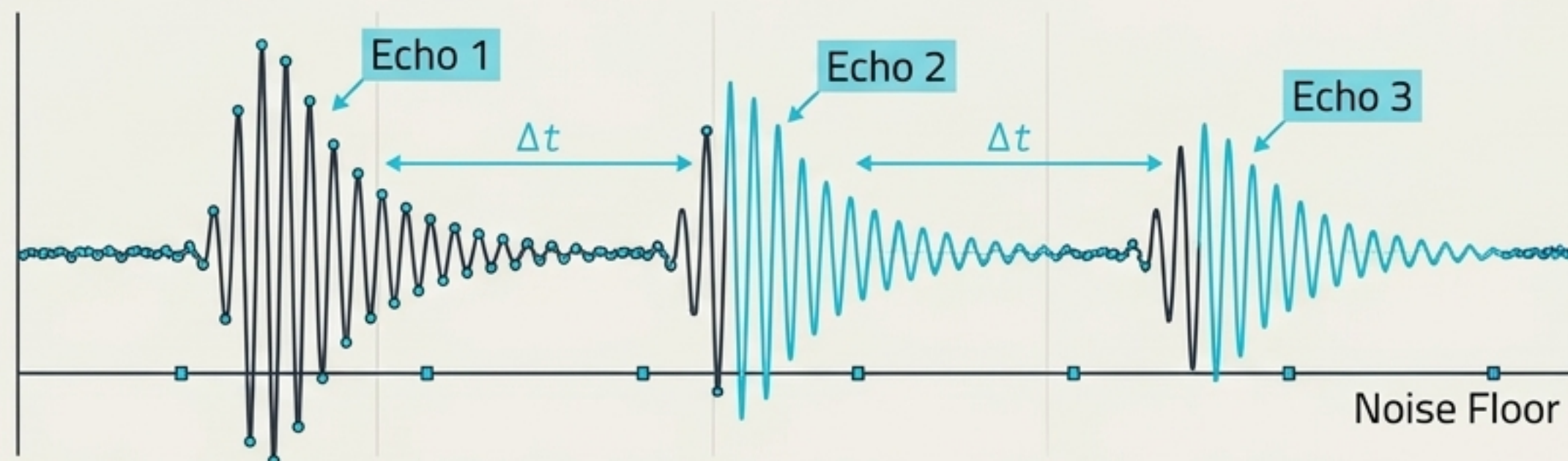
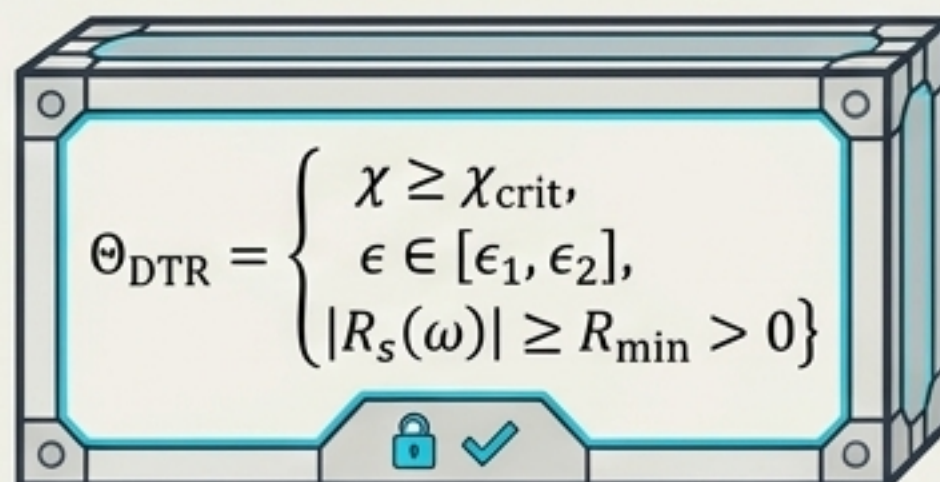
The Falsifiability Mandate

A model that retroactively adjusts its reflectivity down to zero after every null LIGO result is **unfalsifiable**. The DTR hypothesis mandates a bounded, pre-committed parameter space ($R_{\min} > 0$).

Unfalsifiable / Unbounded Model

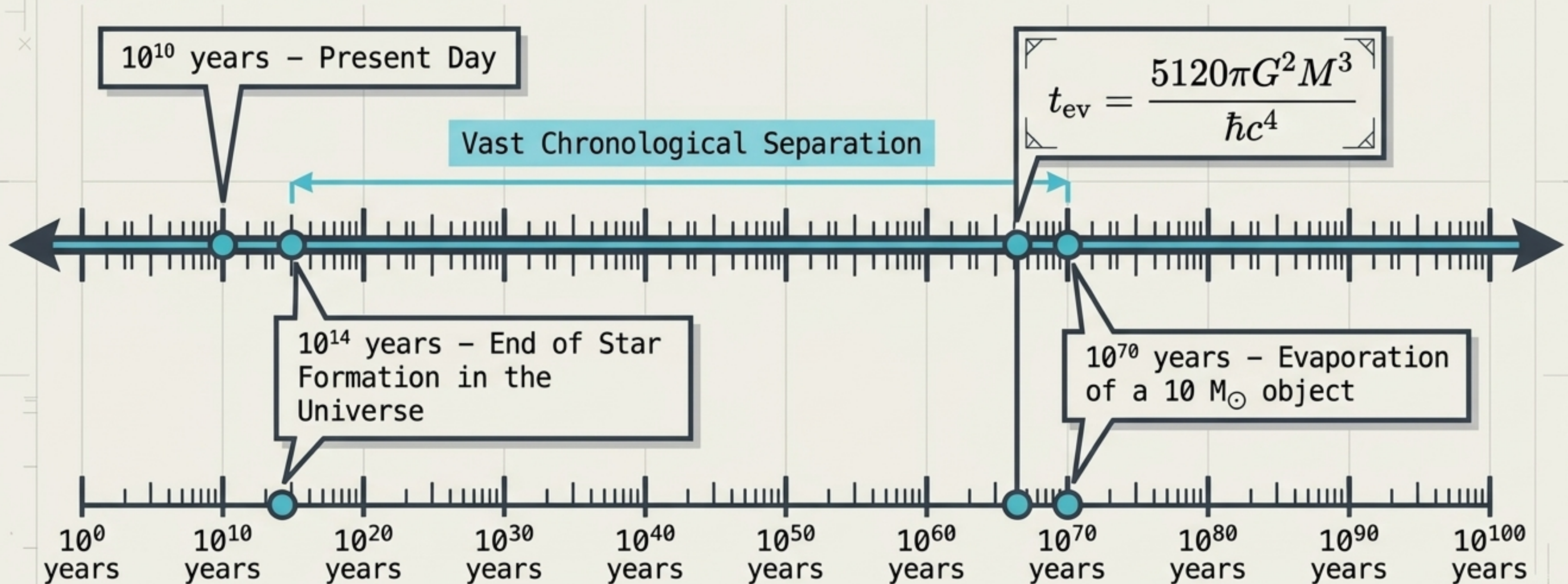


Falsifiable / Bounded Model



Terminal Evaporation: Resolving the Timescale Fallacy

Hawking radiation is an exterior semiclassical effect dictating the far-future of *any* sufficiently compact remnant. Trillions of years marks the end of star formation, not evaporation. The timescale is vastly separated from collapse.



The Burden of Proof

We present a causally connected structure, a defined numerical target, and an observational mandate. The hypothesis is specific enough to be computationally validated—or observationally refused.

